Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: PixelTrendAnalytics
b. Team leader name: Muthukumar Arunachalam
c. Problem Statement: Trace quality evidence into a reviewed inspection plan.

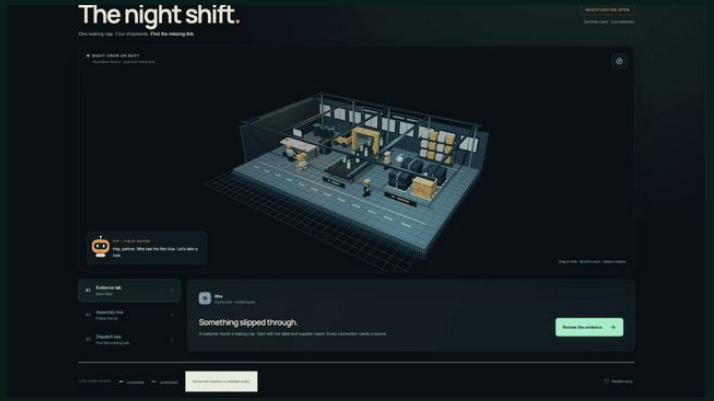
## Brief about the idea

# One leaking cap.
# Possible causes.
# One next check.

Pip investigates.

Operators test and review.

Gemini reassesses.




Synthetic case. Real Gemini. Human-reviewed decisions.

Opportunities

DIFFERENCE: Gemini revises hypotheses after operator results.
SOLUTION: Evidence, inspection missions and reviewed actions.
USP: A playable factory with an auditable agent journal.

# An investigation that learns from results.

Pip proposes missions and revises evidence-backed hypotheses.

Operators approve checks, facts and the final handover.

List of features offered by the solution



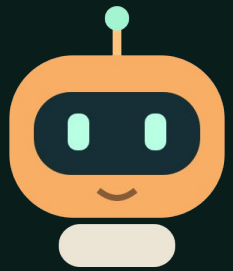
## Mira’s evidence lab

Read evidence. Compare causes. Approve a demo inspection mission and record its result.



## Sol’s assembly line

Review assembly records. Compare inspection scopes with explicit workload assumptions.



## Pip’s investigation

Reassess results with Gemini. Ask operational questions. Review corrective-action drafts.

Process flow diagram or Use-case diagram

## 01 INVESTIGATE
### Compare causes

Read evidence and procedures

Find gaps and contradictions

Propose inspection missions

→

## 02 INSPECT
### Operator checkpoint

Approve a demo mission

Record a synthetic result

Gemini revises its findings

→

## 03 HAND OVER
### Review next steps

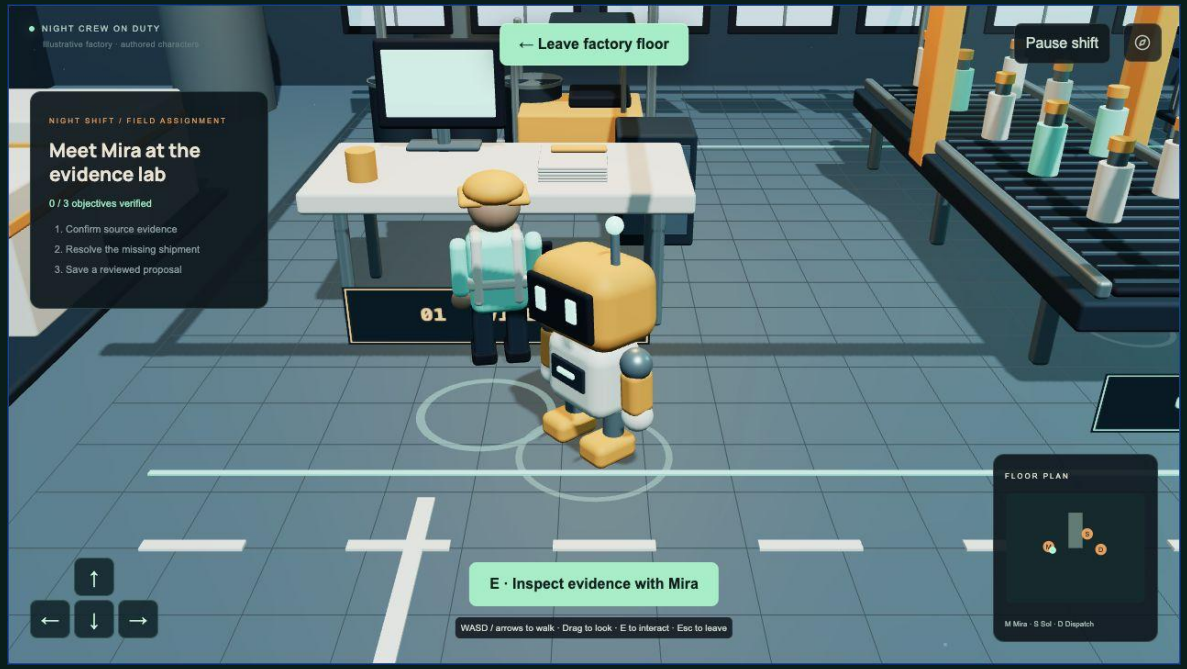
Compare inspection scopes

Approve action drafts

Export a reviewed snapshot

**Operational chat clarifies the case using cited evidence.**
**Humans approve decisions. Demo results are explicitly synthetic.**

# Wireframes/Mock diagrams of the proposed solution (optional)




Walk as Pip. Approach a station. Press E to enter the real evidence-review workflow.

Architecture diagram of the proposed solution




Operations-only Gemini chat uses a separate scope gate and has no mutation tools.

## Technologies to be used in the solution

Gemini 3.5 Flash-Lite via Vertex AI
Tool calling, cause comparison and grounded operational chat.

Cloud Run + Firestore
Bounded execution, mission results and persistent case journals.

React, TypeScript, Three.js + FastAPI
Playable factory, human approvals and deterministic scope.

# Estimated implementation cost (optional)

**ONE MEASURED CASE**

| | TOKENS | RATE / 1M | COST |
|---|---|---|---|
| Input | 66,308 | $0.30 | $0.0199 |
| Output | 3,087 | $2.50 | $0.0077 |
| **Total (19 calls)** | **69,395** | | **$0.0276** |

Gemini 3.5 Flash-Lite on Vertex AI
Investigation, scans, reassessment and 3 chat questions

**AI COST PER CASE**

**≈ $0.03**

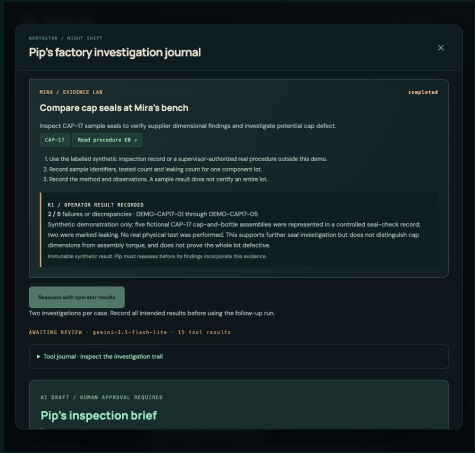
AI inference per complete investigation

**MONTHLY COST: GEMINI AND TOTAL CLOUD BUDGET**

| $2.76 | $27.61 | $6 | $40 |
|---|---|---|---|
| 100 cases<br>Gemini only | 1,000 cases<br>Gemini only | 100 cases<br>Total cloud budget | 1,000 cases<br>Total cloud budget |

Measured synthetic case, 18 Oct 2026. USD before credits and taxes. Total cloud budgets include headroom.
Infrastructure is estimated. 1,000 cases/month requires a higher daily model-call limit.

# Snapshots of the prototype




**Pip proposes.**
**Operators inspect.**

**Gemini reassesses.**
**Humans decide.**

Cause comparisons, inspection missions and operational chat with real Gemini.

## Prototype Performance report/Benchmarking

# 1,500

units proposed for inspection

Zero unresolved after review

ADAPTIVE QUALITY INVESTIGATION

Cause comparison and missions

22 backend tests passed

Human approval gates verified

Synthetic case and operator-entered demo results. No production accuracy claim.

Gemini recommendations cannot approve scope, release stock or operate equipment.

Additional Details/Future Development (if any)

# Next shift: a real-team pilot

Can a quality team reach a defensible decision with less searching and fewer missed connections?

Measure evidence coverage, review time and decision errors.

Next: approved real procedures, controlled uploads and ERP integration.

Try BatchLens and inspect the source

- GitHub: https://github.com/mk-tdev/BatchLens
- Demo video: https://youtu.be/aosveR1UzOM
- Live: https://batchlens-hk3hxwi5na-as.a.run.app/

# Walk the factory.
# Complete a mission.
# Challenge Pip.

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# Thank you

